1. A cosmic ray moving at $\beta = 0.99995$ travels 60 km through the earth's atmosphere, as measured by experimenters on the ground. How long does it take according to the cosmic ray?

2. At what speed does a moving clock tick at half the rate of an identical clock at rest?

3. An astronaut travels to a star system 4.5 ly away at a speed of of 0.90 c. Assume that the time needed to accelerate and decelerate is negligible.

- **A.** How long does the journey take according to mission control on Earth?
- **B.** How long does the journey take according to the astronaut?
- **C.** How much time elapses between the launch and the arrival of the first radio message from the astronaut saying that she arrived?

4. Jill claims her new rocket is 100 m long. As she flies past your house, you measure the rocket's length and find that it is only 80m. What is Jill's speed, as a fraction of c?

5. Our Milky Way galaxy is 100,000 ly in diameter. A spaceship crossing the galaxy measures the galaxy's diameter to be a mere 1.0 ly.

- **A.** What is the spacecraft's speed, as a fraction of c, relative to the galaxy?
- **B.** How long is the crossing time as measured in the galaxy's reference frame?

6. (Extra Credit)The binomial approximation says that if x « 1, then you can rewrite any binomial as:

$$(1+x)^n \approx 1 + nx$$

Use the binomial theorem to help you determine at which speed, in m/s, would a moving clock loose 1.0 ns in 1.0 day according to experimenters on the ground